



MACHINE LEARNING

“Personal Habit Drift Detection & Forecasting System”



Presented by:
Jannat Akther Chaity — 232-115-215
Saima Sultana — 232-115-231



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PROJECT OVERVIEW



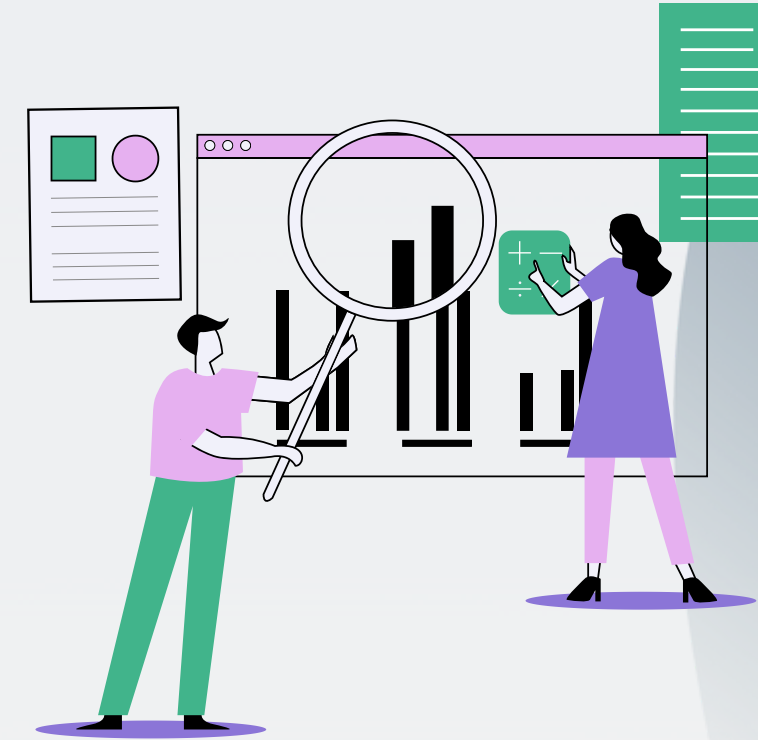
CONCEPT

Our system analyzes daily behavioral patterns to detect habit changes, predict future habit drift using machine learning, and provide early warnings with personalized insights.



OBJECTIVE

- ✓ **Analyze daily behavioral patterns**
- ✓ **Calculate Habit Score & Drift Score**
- ✓ **Detect habit changes**
- ✓ **Predict future drift using ML models**
- ✓ **Provide early warnings**
- ✓ **Generate personalized insights**



WHY THIS SYSTEM?

The Problem:

People's daily habits can gradually change without being noticed.

Examples:

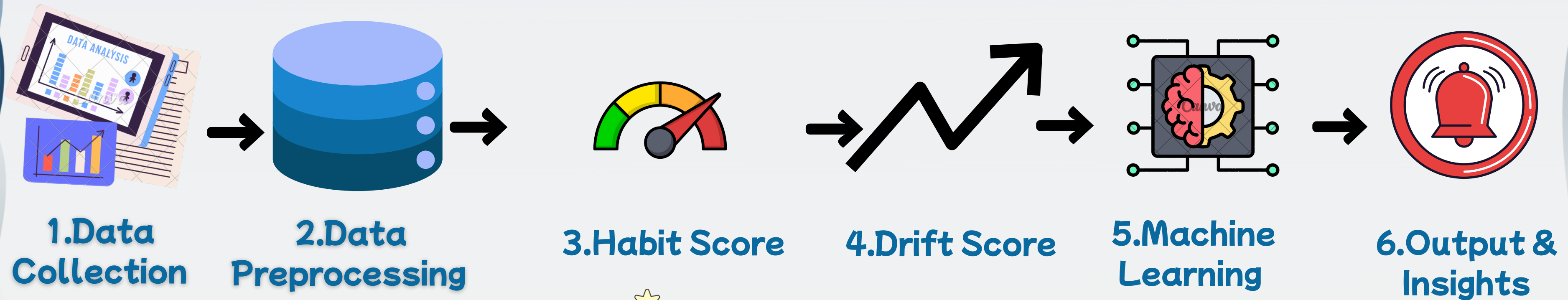
- 😴 Sleep ↓
- 📖 Study Time ↓
- 📱 Screen Time ↑
- 🏃 Exercise ↓
- 😞 Stress ↑

The Challenge:

- Traditional habit tracking mainly shows what happened.
- Our system moves from simple tracking to predictive behavioral analysis.



METHODOLOGY & WORKFLOW



HOW IT WORKS

① Data Collection

Daily behavioral information

② Data

Clean and prepare data

③ Habit Score

Measure daily habit quality

④ Drift Score

Measure change over time

⑤ ML Prediction

Predict future drift

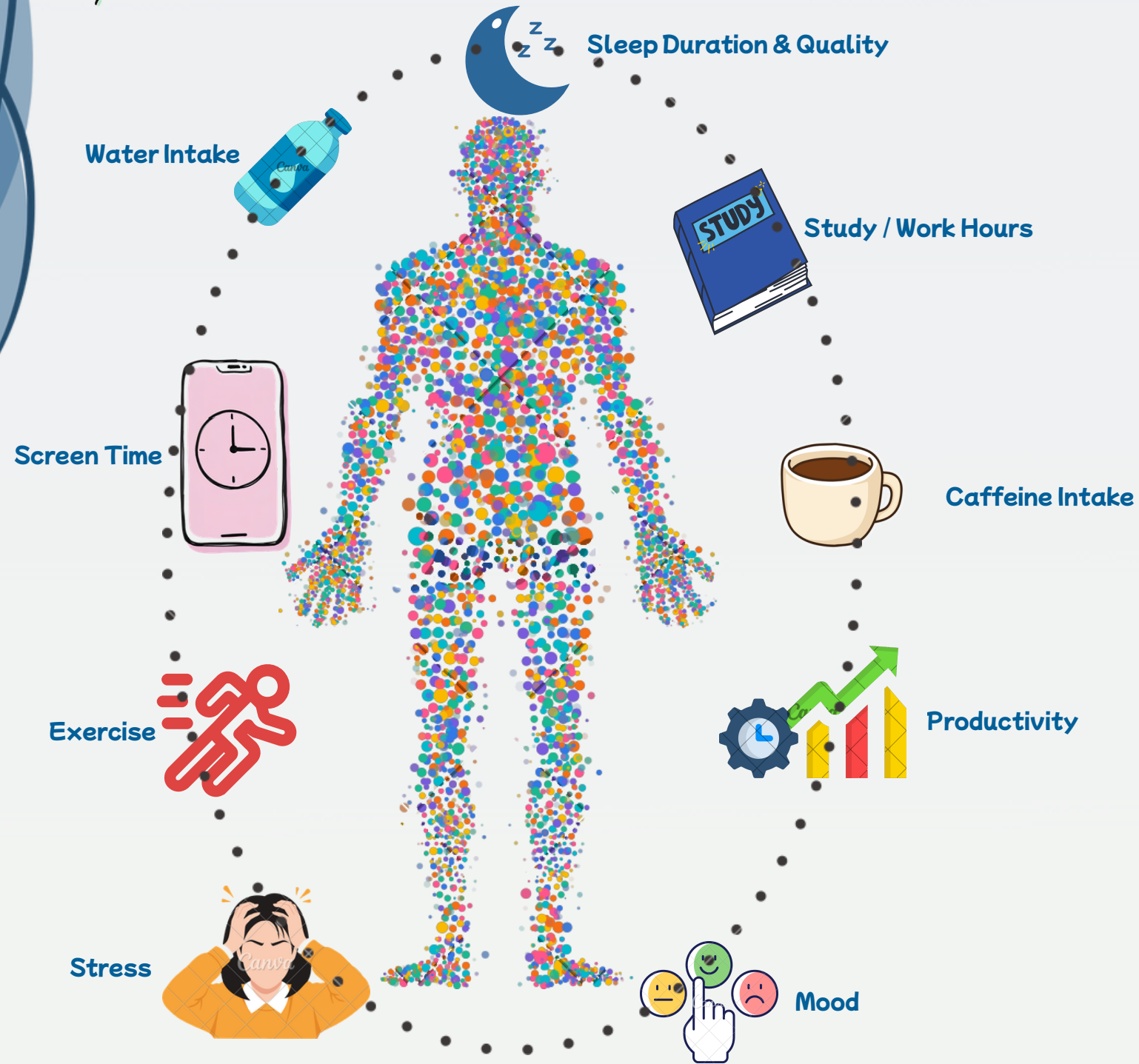
⑥ Output & Insights

Forecast + warnings + insights

DATASET INFORMATION



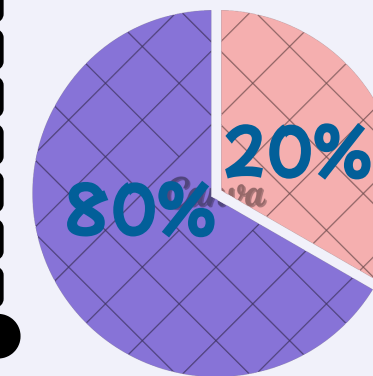
DAILY BEHAVIORAL DATA



DATASET SUMMARY

- **Features:** 10 behavioral features
- **Target Variable:** Drift Score
- **Source:** Collected via Google Form
- **Preprocessed & cleaned for analysis**

DATA SPLIT



- 80% Training Data
- 20% Testing Data



TARGET VARIABLE = Drift Score (Continuous) – Measures change in habit quality from previous day.

HABIT SCORE & DRIFT SCORE

HABIT SCORE (0-100)



The Habit Score represents the quality of daily habits.
Higher score = healthier habits.

DRIFT SCORE



Drift Score shows how much today's Habit Score has changed compared to yesterday.

$$\text{Drift Score} = \text{Today's Habit Score} - \text{Yesterday's Habit Score}$$

EXAMPLES

Decline Example

Yesterday

80

Today

60

$$\text{Drift} = 60 - 80 = -20$$

Declining

Improve Example

Yesterday

60

Today

75

$$\text{Drift} = 75 - 60 = +15$$

Improving

DRIFT CLASSIFICATION



Positive (> 0)
Improving



Near Zero ($= 0$)
Stable



Negative (< 0)
Declining

DRIFT CLASSIFICATION & PERSONAL BASELINE

Drift Classification

Drift Score	Status
Positive	Improving
Near Zero	Stable
Negative	Declining

PERSONAL BASELINE:

The system compares current behavior with the user's own historical pattern.

Example:

Normal Study Time: 6 hrs
Recent Study Time: 3.5 hrs
→ Below personal baseline

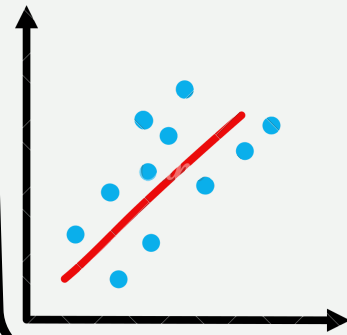


Key idea: Personalized analysis instead of one-size-fits-all comparison.

MACHINE LEARNING MODELS & RESULTS

MODELS USED

Linear Regression



A simple linear approach to predict drift score.

Random Forest Regression



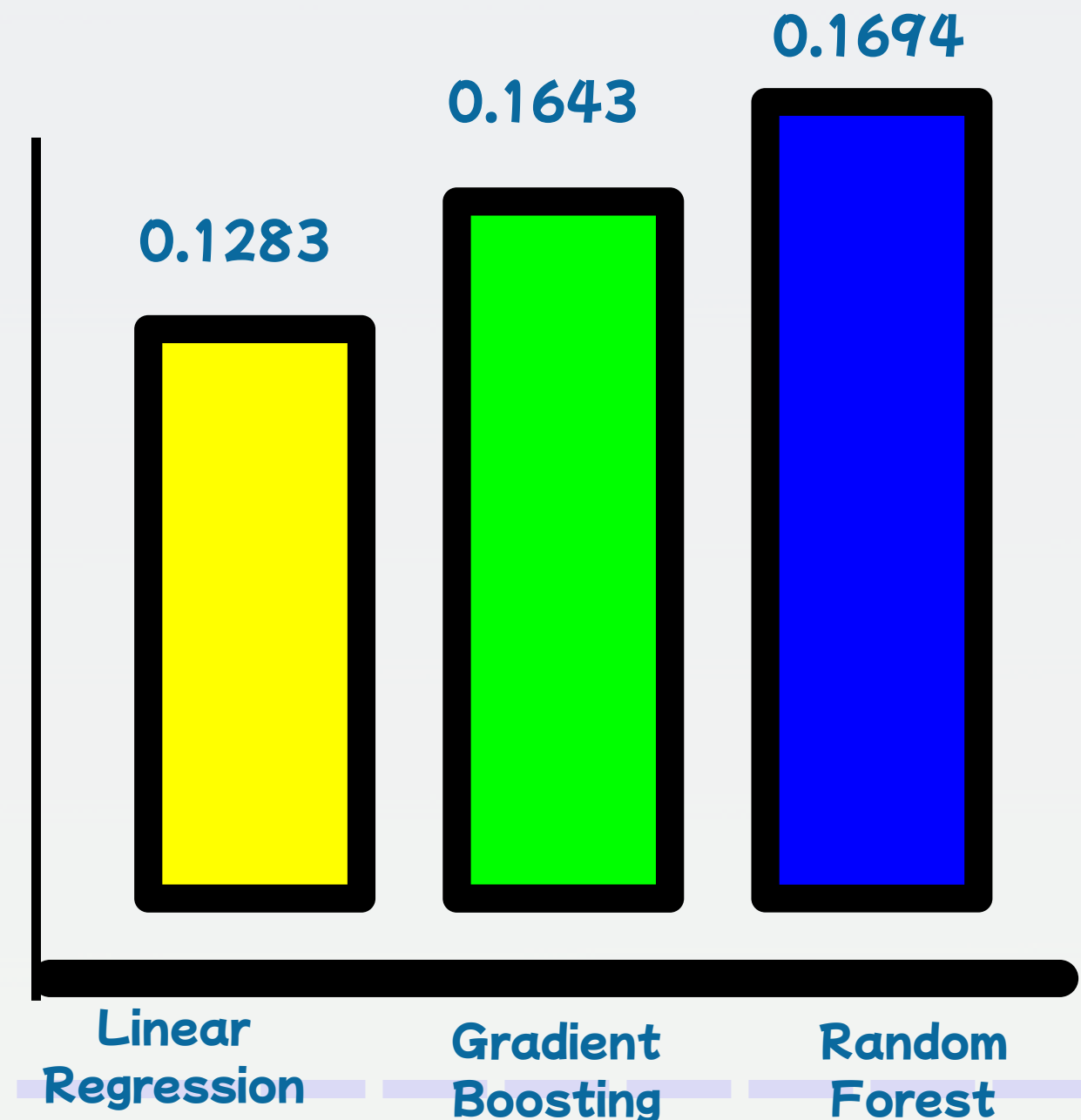
An ensemble of decision trees that captures complex patterns.

Gradient Boosting Regression



Sequential ensemble that corrects previous errors.

MODEL PERFORMANCE (R^2 Score)



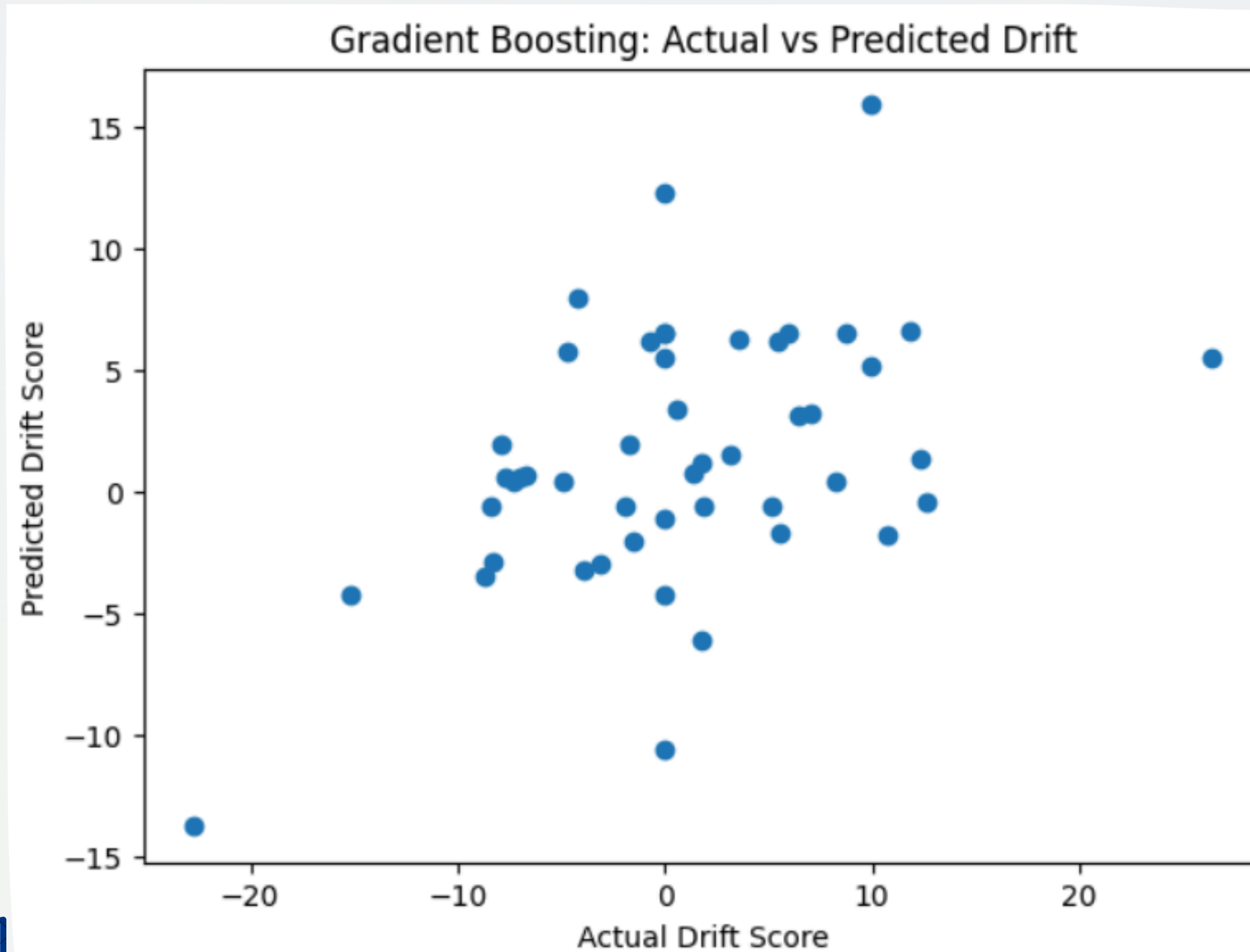
Final Model: Gradient Boosting



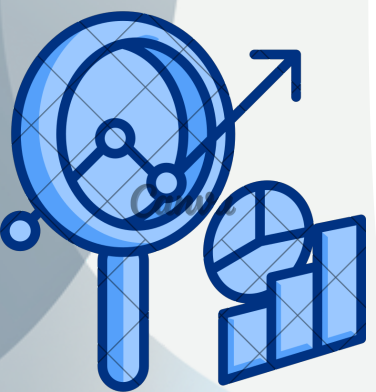
Why Gradient Boosting?

Strong predictive performance + suitable for modeling non-linear relationships between behavioral factors and habit drift.

ACTUAL vs PREDICTED DRIFT SCORE



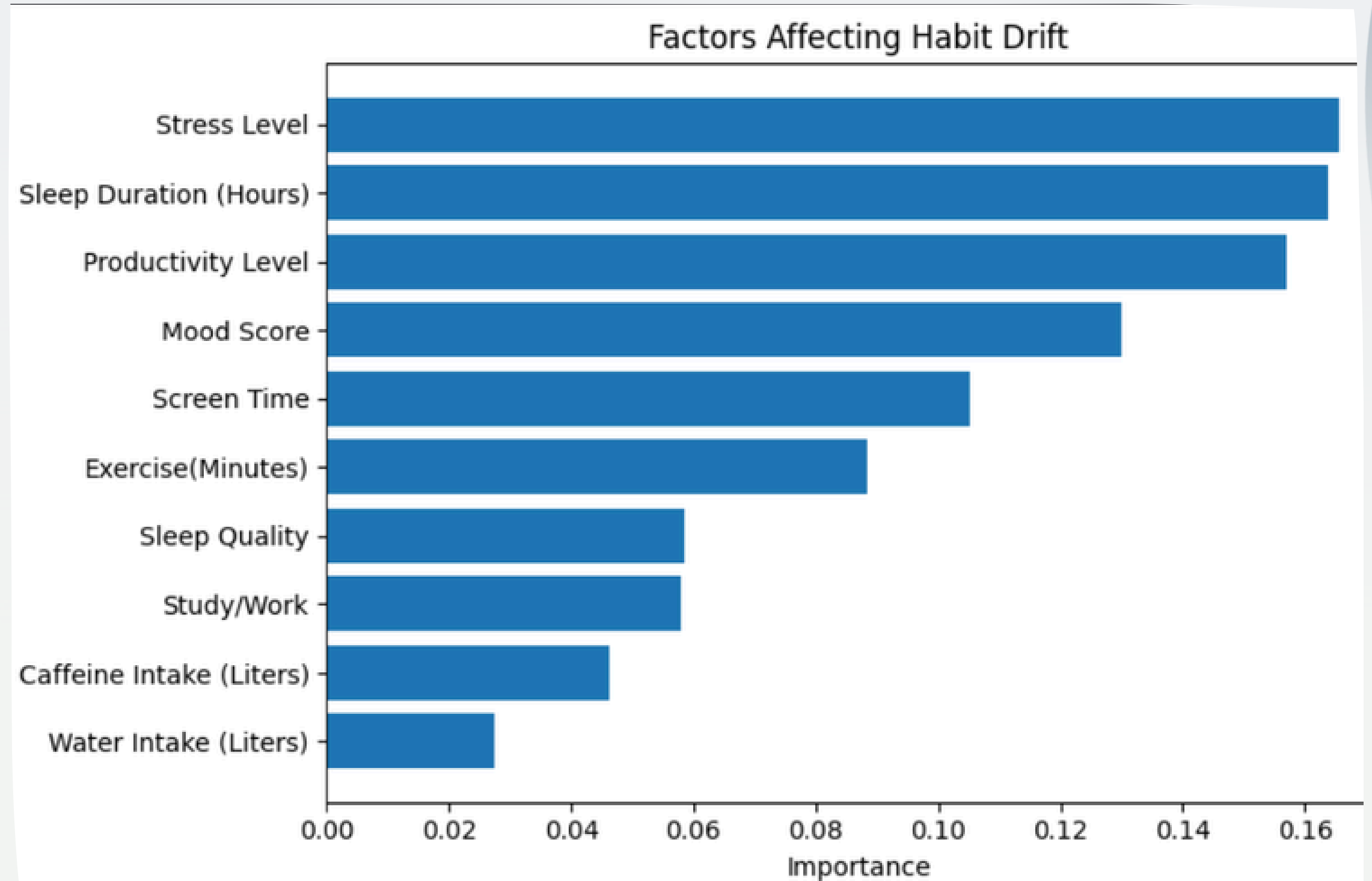
Compares actual Drift Scores with predictions generated by the selected Gradient Boosting model.



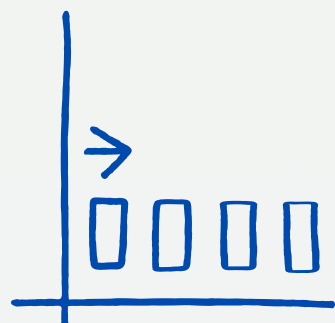
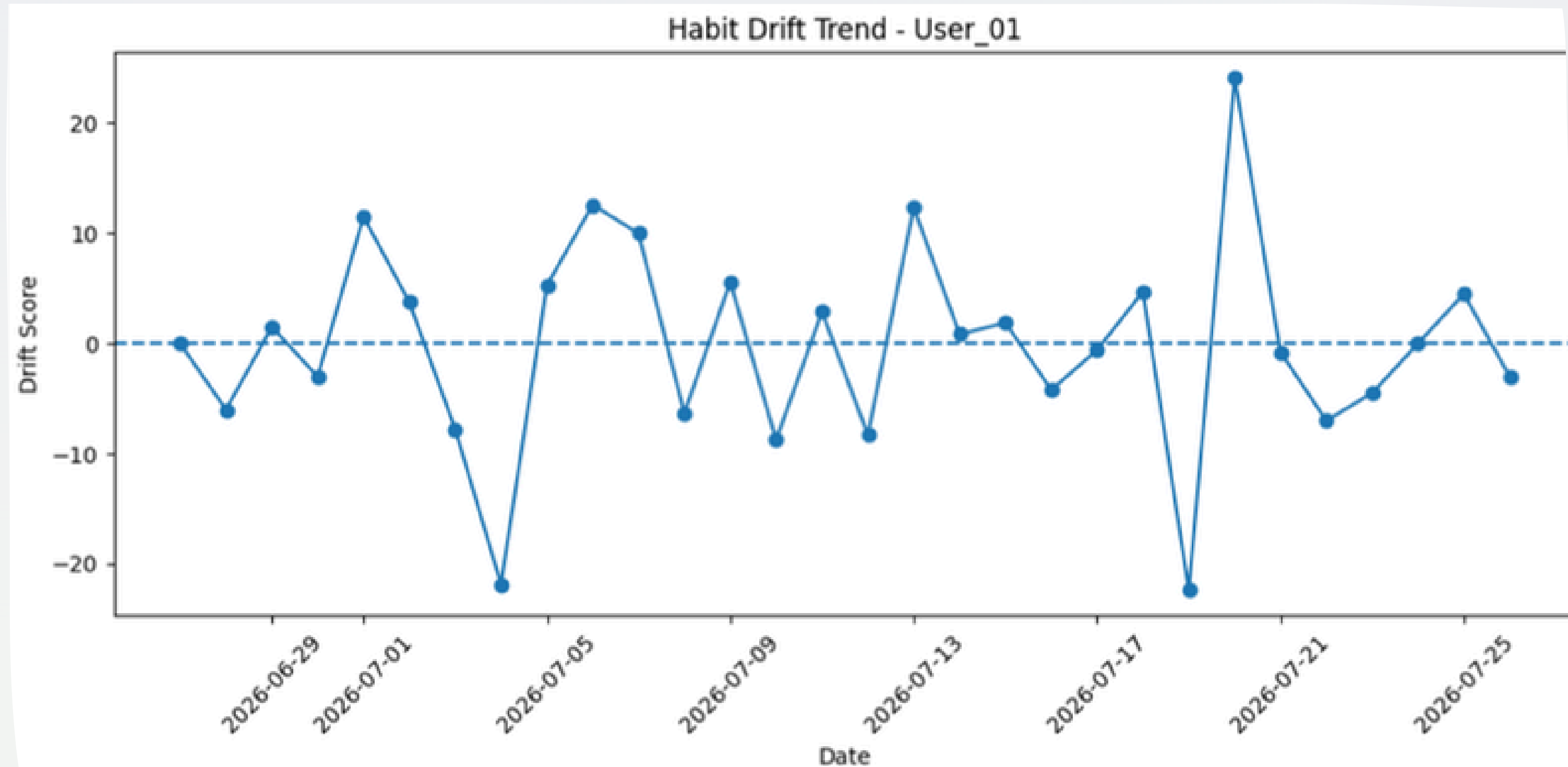
FEATURE IMPORTANCE



Shows which behavioral features contributed more strongly to Drift Score prediction.



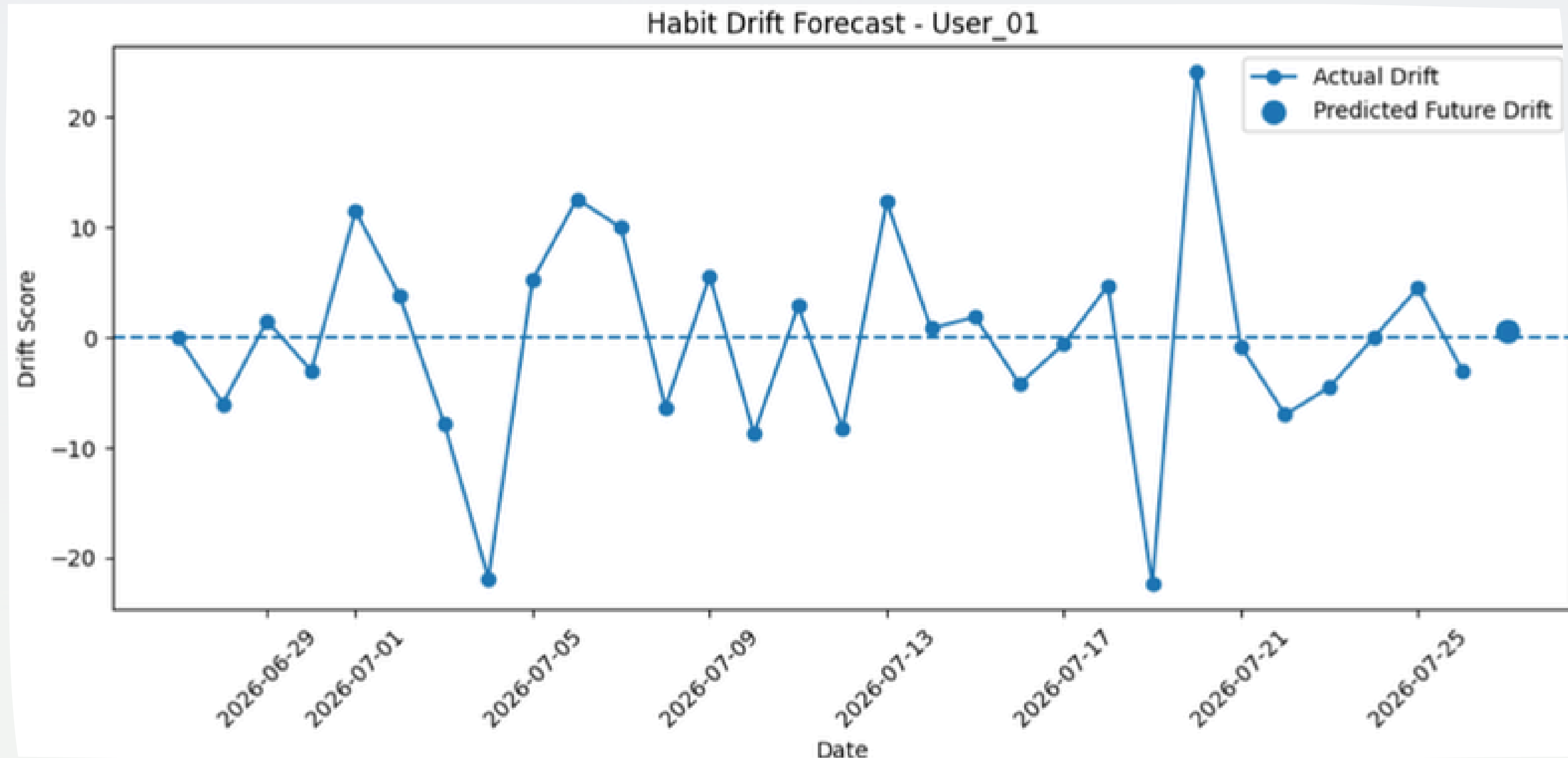
PERSONAL HABIT DRIFT TREND



Tracks changes in the user's Drift Score over time.

Positive → Improvement | Negative → Decline

FUTURE HABIT DRIFT FORECAST



Uses recent Drift Score observations to estimate the next behavioral trend.
Forecast → Improving / Stable / Declining

OUTPUT, INSIGHTS & IMPACT

YOUR HABIT STATUS

DECLINING



Drift Score

-12

WHY?

-  Screen Time ↑
-  Study Time ↓
-  Sleep ↓
-  Exercise ↓

FORECAST



Predicts future drift trend (Improving / Stable / Declining).

EARLY WARNING



Detects repeated negative changes and sends warnings.

PERSONALIZED INSIGHT



Identifies behavioral factors contributing to recent habit changes.

CONCLUSION & FUTURE WORK

CONCLUSION

- ✓ **Daily behavioral analysis**
- ✓ **Habit & Drift Score calculation**
- ✓ **Personal baseline comparison**
- ✓ **ML-based drift prediction**
- ✓ **Future forecasting**
- ✓ **Early warnings**
- ✓ **Personalized insights**



FUTURE WORK



- **Larger datasets**
- **Longer behavioral histories**
- **Time-series models such as LSTM**
- **SHAP-based explainability**
- **Wearable/mobile data integration**
- **Real-time monitoring**

THANK YOU!

From daily data to personalized behavioral insights.

Any
Question?

